

Executive Summary — Mouse USV ASD Classification

XGBoost Binary Classifier | Generated March 28, 2026

OBJECTIVE

Classify mouse pups as **Wild-Type (WT)** or **ASD-model** based on ultrasonic vocalization (USV) features extracted from audio recordings.

DATASET

- **7,322** syllable-sequence samples from aggregated recordings
- **46 features** per sample (syllable frequencies, durations, distances, ISI, sex, age, strain)
- Split: 60% train / 20% validation / 20% test
- Class ratio in test: 473 WT vs 992 ASD (~1:2)

KEY RESULTS

76.0%

Test Accuracy

0.77

Weighted F1-Score

79.8%

Train Accuracy

PER-CLASS PERFORMANCE

Class	Precision	Recall	F1-Score	Support
0 — WT	0.58	0.92	0.71	473
1 — ASD	0.94	0.69	0.80	992
Weighted Avg	0.83	0.76	0.77	1,465

CONFUSION MATRIX

	Predicted WT	Predicted ASD
Actual WT	433	40
Actual ASD	311	681

STRENGTHS

- High WT recall (92%) — almost all healthy mice are correctly identified.
- High ASD precision (94%) — when the model predicts ASD, it is almost always correct.
- Small train-test gap (79.8% vs 76.0%) — no significant overfitting.

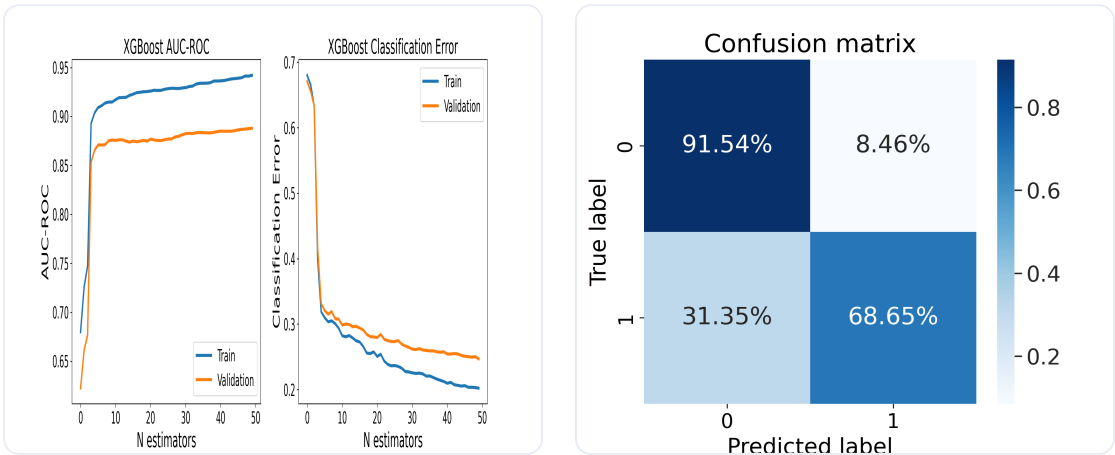
WEAKNESSES

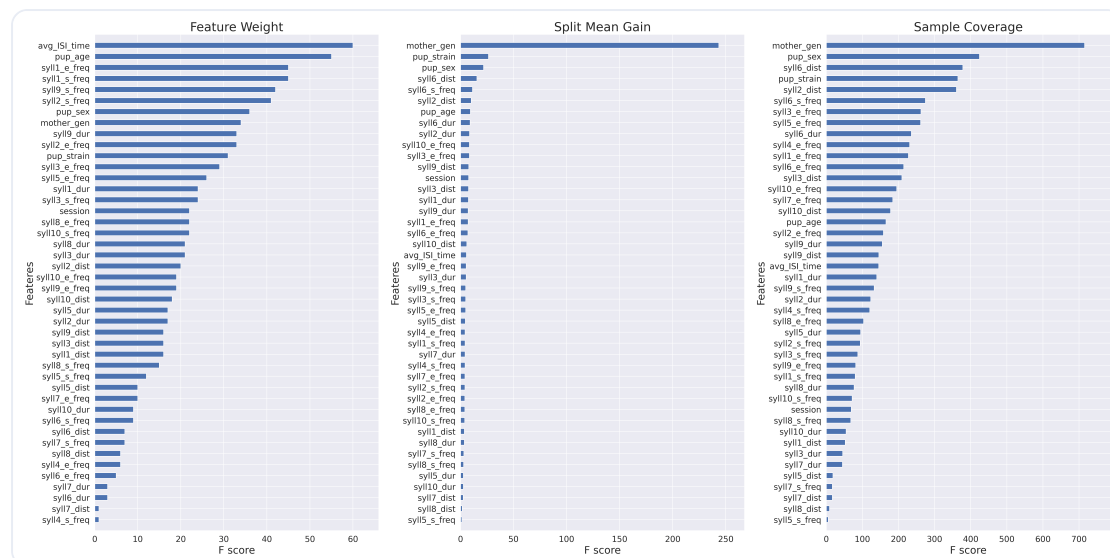
- ASD recall is only 69% — the model misses ~31% of ASD mice (311 false negatives).
- WT precision is low (58%) — many predicted-WT samples are actually ASD.
- Class imbalance (1:2) may bias the model toward predicting WT too often.

TOP FEATURE

The single most important feature is **avg_ISI_time** (average inter-syllable interval), contributing **46%** of total feature importance. This suggests that vocalization timing patterns are the strongest differentiator between WT and ASD mice.

TRAINING CURVES & FEATURE IMPORTANCE





CONCLUSION

The model achieves reasonable classification performance (76% accuracy, 0.77 weighted F1) for a biological signal task. The main limitation is the **31% miss-rate on ASD mice**. For screening purposes where missing ASD cases is costly, the recall on class 1 should be improved — possible directions include threshold tuning, feature engineering on temporal patterns, or ensemble methods.